

MODULE 01 · DOMAIN 1 - AI & ML FUNDAMENTALS

AI & ML Foundations

The vocabulary the rest of the exam assumes you already have.

DOMAIN 1 of 5 EXAM WEIGHT 20% MODULE 1 of 12

The exam rarely asks **what is machine learning**. It describes a business situation and asks **which kind of problem is this**.

AI, ML, deep learning, generative AI

- ◆ **Artificial intelligence** is the outer box. Any system that performs tasks needing human-like reasoning.
- ◆ **Machine learning** sits inside it. Systems that learn patterns from data rather than following written rules.
- ◆ **Deep learning** sits inside ML. Neural networks with many layers, good at images, audio and language.
- ◆ **Generative AI** sits inside deep learning. Models that produce new content rather than a label or a number.

ON THE EXAM

These are nested, not parallel. Generative AI is a kind of deep learning, which is a kind of machine learning, which is a kind of AI.

Know these four by their examples

TYPE	WHAT IT LOOKS LIKE
Structured	Rows and columns with a fixed schema. A CSV of measurements, a database table.
Semi-structured	Tagged but flexible. JSON, XML, log files.
Unstructured	No schema at all. Documents, images, audio, video, free text.
Time-series	Measurements stamped in order. Sensor readings, sales by day.

ON THE EXAM

The classic question gives you four options and asks which is structured data. The answer is the CSV or the database table.

Labeled or unlabeled?

LABELED DATA

You know the right answer

- Each example carries its correct output
- Photos tagged cat or dog
- Transactions marked fraud or legitimate
- Expensive and slow to produce
- Enables supervised learning

UNLABELED DATA

You only have the inputs

- Raw examples with no correct output attached
- A pile of customer records
- Images with nothing marked on them
- Cheap and usually abundant
- Enables unsupervised learning

THE TIEBREAKER

Scenario says the data is mostly unlabeled? The question is heading toward either unsupervised learning or a labeling service like SageMaker Ground Truth.

Supervised or unsupervised?

SUPERVISED

Learn a mapping from labels

- Needs labeled training data
- Classification: predict a category
- Regression: predict a number
- You can measure accuracy directly
- Fraud detection, churn prediction, pricing

UNSUPERVISED

Find structure without labels

- No labels required
- Clustering: group similar items
- Anomaly detection: find the odd ones
- Dimensionality reduction: compress features
- Customer segmentation, image region discovery

THE TIEBREAKER

No labeled data plus a need to group or segment means clustering. That combination appears repeatedly.

Match the scenario to the technique

THE SCENARIO SAYS	THE TECHNIQUE IS
Predict which category this belongs to	Classification
Predict a numeric value or amount	Regression
Group similar records, no labels given	Clustering
Find the unusual or suspicious records	Anomaly detection
Suggest items this user might like	Recommendation
Predict future values from history	Forecasting

ON THE EXAM

Predicting house prices is regression. Predicting whether a loan defaults is classification. The word price or amount means regression.

Reinforcement learning

- ◆ An agent takes actions in an environment and receives **rewards** or penalties.
- ◆ No labeled dataset. The model learns from consequences over many attempts.
- ◆ Fits sequential decisions: robotics, game playing, resource allocation.
- ◆ **RLHF** applies this to language models, using human preferences as the reward signal.

ON THE EXAM

If a scenario mentions rewards, trial and error, or aligning a model to human preferences, reinforcement learning is in play.

The order matters, and the exam tests the order

01

Frame the problem

Define the business goal and whether ML is the right tool.

02

Collect & prepare

Gather data, explore it, clean it, engineer features.

03

Train

Select an algorithm, fit it, tune hyperparameters.

04

Evaluate

Measure against held-out data using a fit-for-purpose metric.

05

Deploy

Put the model behind an endpoint or a batch job.

06

Monitor

Watch for drift and degradation. Retrain when needed.

ON THE EXAM

There is an ordering question that asks you to sequence these steps. Business goal always comes first; monitoring always comes last.

Feature engineering is the step people underestimate

- ◆ **Feature engineering** means creating or selecting the input variables the model trains on.
- ◆ Combining, transforming, or deriving fields. Turning a date into day-of-week and season.
- ◆ **Exploratory data analysis** comes first: understand distributions and spot problems.
- ◆ Better features usually beat a better algorithm. The exam reflects this.

ON THE EXAM

Asked how to improve prediction accuracy on tabular data, the answer that mentions creating or selecting relevant features is usually correct.

Split the data, and never train on the test set

- ◆ **Training set** — the model learns from this.
- ◆ **Validation set** — used to tune hyperparameters and compare candidates.
- ◆ **Test set** — touched once, at the end, to estimate real-world performance.
- ◆ Evaluating on data the model trained on gives a flattering, meaningless score.

Overfitting or underfitting?

OVERFITTING

Memorised the training data

- High accuracy on training data
- Poor accuracy on new data
- Model is too complex for the data
- Fix: more data, simpler model, regularisation
- Fix: stop training earlier

UNDERFITTING

Never learned the pattern

- Poor accuracy on training data
- Poor accuracy on new data too
- Model is too simple for the problem
- Fix: more complex model, better features
- Fix: train longer

THE TIEBREAKER

Read the two numbers. Training high and test low means overfitting. Both low means underfitting.

Bias and variance in one slide

- ◆ **Bias** — the model's assumptions are too rigid. It misses real structure. Causes underfitting.
- ◆ **Variance** — the model reacts to noise in the training data. Causes overfitting.
- ◆ Reducing one tends to increase the other. Good models balance them.
- ◆ This statistical bias is **not** the same as societal bias in Domain 4. Same word, different topic.

ON THE EXAM

The word bias means two different things on this exam. Read the surrounding sentence to work out which one is meant.

What each number actually tells you

METRIC	ANSWERS THE QUESTION
Accuracy	What fraction of all predictions were right? Misleading on imbalanced data.
Precision	Of the ones we flagged, how many were actually positive?
Recall	Of the ones that were actually positive, how many did we catch?
F1 score	A single number balancing precision and recall.
AUC-ROC	How well does the model separate the two classes across all thresholds?
Confusion matrix	The raw counts: true and false, positive and negative.

ON THE EXAM

Accuracy is the trap answer when classes are imbalanced. A model that always predicts 'not fraud' scores 99% accuracy and catches nothing.

Precision or recall?

OPTIMISE PRECISION WHEN

False alarms are costly

- Acting on a wrong flag causes real harm
- Blocking a legitimate customer transaction
- Sending an accusatory notice in error
- You would rather miss some than be wrong

OPTIMISE RECALL WHEN

Misses are costly

- Failing to catch a case causes real harm
- Missing a disease in a screening test
- Letting fraud through undetected
- You would rather over-flag than miss

THE TIEBREAKER

Ask which mistake hurts more in the scenario. False alarm hurts more, choose precision. Missed case hurts more, choose recall.

When the output is a number

METRIC	READS AS
MSE	Average squared error. Punishes large mistakes heavily.
RMSE	Square root of MSE, back in the units of the thing you predicted.
MAE	Average absolute error. Less sensitive to outliers than MSE.
R-squared	How much of the variation the model explains. Closer to 1 is better.

ON THE EXAM

Classification metrics on a regression problem, or the reverse, is a common wrong answer. Check whether the output is a category or a number first.

The models that predict best are usually the models that explain themselves **worst**.

Ranked from readable to opaque

MODEL	INTERPRETABILITY
Linear regression	Highest. Each coefficient is a readable weight.
Decision trees	High. You can follow the branch logic to the outcome.
Random forests	Moderate. Feature importance, but no single readable path.
Neural networks	Low. Accurate, but the internal logic is not directly readable.

ON THE EXAM

Asked which technique offers the MOST explainability, the answer is linear regression.
Asked which documents its inner mechanism, decision trees.

How you explain a model you cannot read

- ◆ **Partial dependence plots** show how changing one feature moves the prediction.
- ◆ **SHAP values** attribute a single prediction to the features that drove it.
- ◆ **Feature importance** ranks which inputs matter most overall.
- ◆ In AWS, **SageMaker Clarify** is the service that produces these, plus bias reports.

ON THE EXAM

A transparency report for stakeholders should contain partial dependence plots — not training code or sample data.

Real-time or batch inference?

REAL - TIME

Answer now, one at a time

- A persistent endpoint waits for requests
- Milliseconds matter
- You pay for the endpoint while it runs
- Fraud checks at checkout, chat responses

BATCH

Answer later, all at once

- A job processes a whole dataset, then stops
- No endpoint to keep running
- Cheaper per prediction
- Weekly scoring, bulk document processing

THE TIEBREAKER

If the scenario says results are not needed immediately, or the work runs on a schedule, batch is the cost-correct answer.

When machine learning is the wrong tool

- ◆ The rules are known, fixed and writable. Use code, not a model.
- ◆ You need the same answer every time, with a guarantee. Models are probabilistic.
- ◆ There is not enough data, and no realistic way to get it.
- ◆ A mistake is unacceptable and no human review is possible.

ON THE EXAM

Some questions offer a deterministic, rules-based solution among the options. When the logic is fully known, that is the correct answer.

A logistics company stores five years of delivery records. Each record has the route, the vehicle type, the weather, and the minutes the delivery took.

The company wants to tell dispatchers how long a new delivery will take before the driver leaves.

Which type of ML problem is this?

- ☐ A Classification
- ☐ C Clustering

- ☐ B Regression
- ☐ D Anomaly detection

◆ PAUSE AND DECIDE

Which type of ML problem is this?

- A

 Classification
Output would be a category, but minutes is a quantity.
- B

 Regression
Labeled history, and the output is a number.
- C

 Clustering
Unsupervised. The data is already labeled with actual times.
- D

 Anomaly detection
Finds unusual deliveries. It does not predict duration.

THE SIGNAL

The output is minutes, which is a number. Number out means regression, every time.

A hospital builds a model that screens chest scans and flags patients who may need urgent review by a radiologist.

Every flagged scan is reviewed by a doctor within the hour. The department has capacity to review far more scans than it currently receives.

Which metric should the team optimise?

- | | |
|-------------------------------------|---|
| ☐ A Accuracy | ☐ B Precision |
| ☐ C Recall | ☐ D Mean squared error |

◆ PAUSE AND DECIDE

Which metric should the team optimise?

- A

 Accuracy
Most scans are normal, so a do-nothing model scores well.
- B

 Precision
Would reduce false alarms, but the cost here is a missed case.
- C

 Recall
Catches the most true cases. Capacity absorbs the false alarms.
- D

 Mean squared error
A regression metric. This is a classification problem.

THE SIGNAL

Two details decide it: a missed case is dangerous, and the department has spare review capacity. Both point at recall.

An insurance company trains a model to predict claim amounts. On the training data it achieves an R-squared of 0.97.

On the held-out test data, R-squared drops to 0.41.

What is happening, and what should the team try first?

- ☐ A Underfitting. Train the model for longer.
- ☐ B Overfitting. Simplify the model or add training data.
- ☐ C Data leakage. Remove the target from the features.
- ☐ D The test set is too small. Merge it into training.

◆ PAUSE AND DECIDE

What is happening, and what should the team try first?

- A

 Underfitting
Training score would also be poor. It is 0.97.
- B

 Overfitting
Excellent on seen data, weak on unseen. Textbook pattern.
- C

 Data leakage
Leakage inflates BOTH scores, not just training.
- D

 Merge test into training
Never. You would lose any honest estimate of performance.

THE SIGNAL

Two numbers, far apart, training higher. That gap is the definition of overfitting.

A bank's loan approval model is a deep neural network. It outperforms the bank's previous model by a wide margin.

A regulator now requires the bank to show, for any individual rejected applicant, which factors drove that decision.

What should the bank do?

- ☐ A Replace the network with a decision tree
- ☐ B Publish the model's training code
- ☐ C Use SHAP values to attribute each prediction
- ☐ D Increase the size of the training dataset

◆ PAUSE AND DECIDE

What should the bank do?

- A

 Swap to a decision tree
Valid, but sacrifices accuracy before trying explainability tools.
- B

 Publish training code
Code is not an explanation of an individual decision.
- C

 SHAP values
Attributes a single prediction to the features that drove it.
- D

 More training data
Improves accuracy. Does nothing for explainability.

THE SIGNAL

The requirement is per-applicant, not overall. SHAP explains one prediction; feature importance only explains the model as a whole.

Carry these into the exam

- ◆ AI contains ML contains deep learning contains generative AI. Nested, not parallel.
- ◆ Category out means classification. Number out means regression. No labels plus grouping means clustering.
- ◆ Training high and test low is overfitting. Both low is underfitting.
- ◆ Accuracy is the trap on imbalanced data. Ask whether a false alarm or a miss hurts more.
- ◆ Most explainable is linear regression. Least is a neural network. Clarify produces the explanations.
- ◆ Not needed immediately means batch inference.